Leveraging Explainable AI for Transparent and Energy-Efficient Cybersecurity in IoT-Enabled Computer Network Systems

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Abstract

The integration of Internet of Things (IoT) technologies into modern network systems has amplified both cybersecurity risks and energy consumption challenges. While artificial intelligence (AI) has significantly improved threat detection capabilities, the lack of transparency in many AI models limits trust, interpretability, and efficient deployment in resource-constrained environments. This study explores the role of explainable artificial intelligence (XAI) in enhancing transparency, accountability, and energy efficiency in IoT cybersecurity systems. A novel framework is proposed that integrates explainable machine learning models with energy-aware optimization techniques to provide interpretable threat detection while minimizing computational overhead. Through simulation-based evaluation, the framework demonstrates improved detection accuracy, reduced false positives, and optimized energy consumption compared to conventional AI-based security systems. The results highlight the potential of XAI to enable trustworthy, efficient, and sustainable cybersecurity solutions for IoT-enabled networks.

Keywords: Explainable AI, IoT security, energy efficiency, cybersecurity, machine learning, transparency, sustainable networks

1. Introduction

The rapid adoption of IoT-enabled computer network systems has transformed the way devices communicate, process data, and deliver services across domains such as smart cities, healthcare, and industrial automation. However, this transformation has introduced significant cybersecurity challenges due to the large number of interconnected devices and their inherent resource constraints. Traditional security mechanisms are often insufficient for handling the scale and complexity of IoT networks, leading to the widespread adoption of AI-based solutions for threat detection and mitigation. While these solutions offer improved accuracy and adaptability, they often function as “black-box” models, making it difficult to understand their decision-making processes.

The lack of transparency in AI models poses critical challenges in cybersecurity applications, where trust, accountability, and compliance are essential. In addition, complex AI models often require significant computational resources, which can increase energy consumption and reduce the operational lifespan of IoT devices. This creates a need for solutions that not only enhance security but also ensure interpretability and energy efficiency.

Explainable AI (XAI) has emerged as a promising approach to address these challenges by providing insights into model decisions and enabling more efficient resource utilization. By making AI systems more transparent, XAI can improve trust and facilitate better decision-making in cybersecurity contexts. This study aims to investigate how XAI can be leveraged to design transparent and energy-efficient cybersecurity systems for IoT networks, offering a comprehensive framework that balances performance, interpretability, and sustainability.

2. Literature Review

2.1 Challenges of Black-Box AI in IoT Cybersecurity

AI-based cybersecurity systems have demonstrated high accuracy in detecting threats; however, their lack of interpretability remains a major limitation. Black-box models, such as deep neural networks, often provide predictions without clear explanations, making it difficult for network administrators to understand or trust their outputs. This lack of transparency can hinder incident response and complicate regulatory compliance. Moreover, the computational complexity of these models increases energy consumption, which is particularly problematic in IoT environments with limited resources.

2.2 Explainable AI Techniques

Explainable AI techniques aim to make model decisions more interpretable without significantly compromising performance. Methods such as feature importance analysis, decision trees, and model-agnostic approaches like SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) have been widely used to provide insights into model behavior. These techniques enable users to identify key factors influencing predictions, improving transparency and trust. In IoT cybersecurity, XAI can help identify the root causes of anomalies and facilitate more effective threat mitigation strategies.

2.3 Energy Efficiency in AI-Driven Security Systems

Energy efficiency is a critical consideration in IoT networks, where devices often operate under strict power constraints. AI models, particularly deep learning systems, can consume significant energy during training and inference. Researchers have explored various approaches to address this issue, including model compression, edge computing, and energy-aware algorithms. The integration of XAI can further enhance energy efficiency by enabling the selection of simpler, more interpretable models that require fewer computational resources. This dual benefit of transparency and efficiency makes XAI a valuable tool for sustainable IoT cybersecurity.

3. Methodology

This study proposes an explainable AI-based cybersecurity framework designed to operate efficiently in IoT-enabled network systems. The framework integrates interpretable machine learning models with energy-aware optimization techniques to achieve transparent and sustainable

threat detection. A simulation-based approach is used to evaluate the performance of the proposed system under various network conditions and attack scenarios.

Parameter	Description
Network Size	500–900 IoT devices
Attack Types	DDoS, intrusion, data manipulation
AI Models	Decision Trees, Gradient Boosting, SHAP
Evaluation Metrics	Accuracy, interpretability score, energy usage
Simulation Environment	Python / NS-3
Optimization Strategy	Model simplification and adaptive processing

The first phase involves training interpretable machine learning models using datasets that include both normal and malicious network traffic. Feature selection techniques are applied to identify the most relevant attributes influencing network behavior. Explainability tools such as SHAP are used to analyze model predictions and provide insights into the decision-making process.

In the second phase, an energy-aware optimization module is incorporated into the framework. This module dynamically adjusts model complexity and processing frequency based on network conditions and device energy levels. By prioritizing critical tasks and reducing unnecessary computations, the system achieves a balance between performance and energy efficiency.

4. Results

The performance of the proposed XAI-based cybersecurity framework is evaluated using key metrics related to detection accuracy, interpretability, and energy consumption.

Model Type	Accuracy (%)	Interpretability Score	Energy Consumption (J)
Black-Box AI Model	95.8	Low	125
Traditional ML Model	89.7	Medium	105
Proposed XAI Framework	94.2	High	88

The results indicate that the proposed XAI framework achieves a strong balance between accuracy and interpretability while significantly reducing energy consumption. Although the black-box model achieves slightly higher accuracy, it lacks transparency and consumes more energy.

Additionally, the interpretability provided by the XAI framework enables better understanding of network behavior and facilitates more effective threat mitigation. This demonstrates the practical value of integrating explainability into AI-driven cybersecurity systems.

5. Discussion

The findings of this study demonstrate that explainable AI can play a pivotal role in addressing the limitations of traditional AI-based cybersecurity systems in IoT environments. By providing insights into model decisions, XAI enhances transparency and trust, which are essential for effective security management. The ability to interpret model outputs allows network administrators to understand the causes of detected anomalies, enabling more informed and timely responses to cyber threats. Furthermore, the use of interpretable models reduces computational complexity, contributing to lower energy consumption and improved system sustainability. This highlights the importance of balancing model performance with interpretability and efficiency in IoT cybersecurity applications.

Another key implication of this research is the potential for integrating XAI with emerging technologies such as edge computing and federated learning. These approaches can further enhance the scalability and privacy of IoT security systems while maintaining energy efficiency. However, challenges remain in terms of standardizing interpretability metrics and ensuring that explanations are meaningful and actionable. Future research should focus on developing unified frameworks that combine explainability, efficiency, and scalability, as well as exploring real-world deployment scenarios. The adoption of XAI-driven cybersecurity solutions represents a significant step toward building secure, transparent, and sustainable IoT network systems.

6. Conclusion

This study has explored the application of explainable AI in designing transparent and energy-efficient cybersecurity systems for IoT-enabled computer networks. The proposed framework demonstrates that XAI can effectively balance detection performance, interpretability, and energy efficiency, addressing key challenges in IoT security. By integrating interpretable models with energy-aware optimization techniques, the system achieves robust threat detection while minimizing resource consumption. Despite the promising results, further research is needed to address challenges related to scalability, real-time implementation, and standardization of explainability methods. Future work should focus on developing lightweight XAI models and exploring their integration with advanced IoT architectures. Ultimately, leveraging explainable AI offers a viable pathway toward achieving secure, efficient, and trustworthy IoT ecosystems.

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